Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Zhiyuan Chen

Department of Statistics
University of Chicago

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Introduction

Literature Review

Methods

Results

References

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Title

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- [1][5-8]

- [2][1-4]
- [2][5-9]
- Results accessible at
  `https://scholar.google.com`

Title

- [3][1-4]

Microsoft® Windows	Apple® Mac OS
Windows-Kernel	Unix-like
Arm, Intel	Intel, Apple Silicon
Sudden update	Stable update
Less security	More security
...	...

Algorithms

Non-Numbering Formula

$$J(\theta) = \mathbb{E}_{\pi_\theta}[G_t] = \sum_{s \in \mathcal{S}} d^\pi(s) V^\pi(s) = \sum_{s \in \mathcal{S}} d^\pi(s) \sum_{a \in \mathcal{A}} \pi_\theta(a|s) Q^\pi(s, a)$$

Multi-Row Formula[1]

$$\begin{aligned} Q_{\text{target}} &= r + \gamma Q^\pi(s', \pi_\theta(s') + \epsilon) \\ \epsilon &\sim \text{clip}(\mathcal{N}(0, \sigma), -c, c) \end{aligned} \tag{1}$$

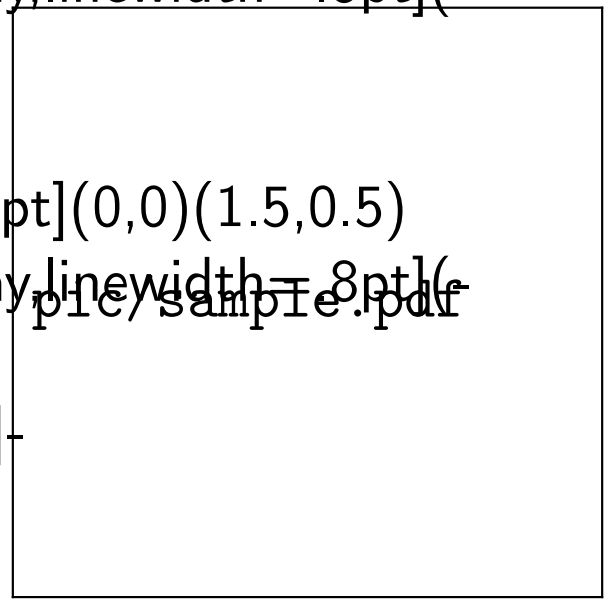
[1] If text appears in the formulause \mathrm{} or \text{} instead

Numbered Multi-line Formula

$$
\begin{aligned}
A = \lim_{n\to\infty} \Delta x \Big(a^2 + \Big(a^2 + 2a\Delta x + (\Delta x)^2\Big) \\
+ \Big(a^2 + 2\cdot 2a\Delta x + 2^2(\Delta x)^2\Big) \\
+ \Big(a^2 + 2\cdot 3a\Delta x + 3^2(\Delta x)^2\Big) \\
+ \dots \\
+ \Big(a^2 + 2\cdot (n-1)a\Delta x + (n-1)^2(\Delta x)^2\Big)\Big) \\
= \frac{1}{3}\left(b^3 - a^3\right) \quad (2)
\end{aligned}
$$

Graphics and Columns

unit=0.8cm (-1.75,-3)(3.25,4) [linewidth=0.25pt](0,0)(0,4)
[tl]0(0.2,2)$\vec{e}_z$ [tr]0(-0.9,1.4)$\vec{e}$ [tl]0(2.8,-1.1)$\vec{C}_{ptmext}$
[br]0(-0.3,2.1)θ
25(0,0)[fillstyle=solid,fillcolor=lightgray,linewidth=.8pt](-0.1,-3.2)(0.1,0)
25(0,0)[fillstyle=solid,fillcolor=yellow,linewidth=3pt](0,0)(1.5,0.5)
25(0,0)[fillstyle=solid,fillcolor=lightgray,linewidth=.8pt](-0.1,0)(0.1,3.2)
25(0,0)[linecolor=red,linewidth=1.5pt]->(0,0)(0.,2)
[linecolor=red,linewidth=1.25pt]->(0,0)(0,2)
[linecolor=red,linewidth=1.25pt]->(0,0)(3,1)



LaTeX Common Commands

Commands

\chapter	\section	\subsection	\paragraph
chapter	section	sub-section	paragraph
\centering	\emph	\verb	\url
center	emphasize	original	hyperlink
\footnote	\item	\caption	\includegraphics
footnote	list item	caption	insert image
\label	\cite	\ref	
label	citation	refer	

Environment

table	figure	equation
table	figure	formula
itemize	enumerate	description
non-numbering item	numbering item	description

LaTeX Examples of environmental commands

[language=TeX]

- A
- B
- C
  - C-1

- A
- B
- C
  - C-1

LaTeX Examples of environmental commands

[language=TeX]

- A
- B
- C
  - C-1

[language=TeX]

1. A
2. B
3. C

n+e

- A
- B
- C
  - C-1

1. A
2. B
3. C

n+e

LaTeX Formulas

[language=TeX] $V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi r^3$

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3 \tag{3}$$

$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi r^3$

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3 \tag{4}$$

- more information here

[language=TeX]

Tabel: numbers meaning

number	meaning
1	4.0
2	3.7

Tabel: numbers & meaning

numbers	meaning
1	4.0
2	3.7

formula (4) at previous slide and Table 2

- [4][1-4]
- [4][5-9]
- [5][1-4]
- [5][5-8]

Thank You